



Course Syllabus for Front-End Web Development with React Training

Presented by:

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COURSE OUTLINE: FRONT-END WEB DEVELOPMENT WITH REACT TRAINING

1.1 Designing with Figma & AI

- Overview of Figma and its features
- Creating a Figma account and navigating the interface
- Understanding Figma's workspace (panels, toolbar, properties)
- Basic drawing tools and shapes
- Grouping, aligning, and arranging elements
- Elements of UI Design: Color, Typography, and Iconography
- Grids and Layout Systems
- Auto Layouts and its use
- Creating reusable components with variants
- Creating interactive prototypes, adding and managing interactions
- Using transitions and animations
- Using auto-layout for responsive design
- Introduction to AI in UI/UX design
- AI-powered design tools (Adobe Sensei, Uizard, Figma AI plugins)

1.2 Introduction to HTML & Web Page Structure

- What is HTML? Role in Web Development
- Structure of an HTML Document
- Basic HTML Elements (Headings, Paragraphs, Links, Lists)
- Images & Figure
- Hands-on: Creating a Simple Web Page

1.3 AI-Assisted HTML Forms & Tables

- Forms & Input Fields (Text, Email, Password, Buttons)
- Form Validation & Attributes
- Creating Tables in HTML
- Using AI for Form Validation & Autocomplete Suggestions
- Hands-on: Building a Registration Form

1.4 Semantic HTML & Best Practices

- Understanding Semantic Tags (article, section, nav, footer)
- SEO-Friendly HTML Markup
- Using AI for SEO Optimization & HTML Cleanup
- Hands-on: Building a Blog Layout with Semantic HTML

1.5 Version Control with Git

- Introduction to version control
- Installing and setting up Git
- Basic Git commands (init, clone, add, commit, push, pull)
- Understanding branches and merging
- Resolving merge conflicts
- Working with remote repositories (GitHub)

1.6 Introduction to CSS & AI in Design

- What is CSS?
- Types of CSS: Inline, Internal, External
- CSS Syntax & Selectors
- CSS Units (px, em, rem, %, vw, vh)
- AI in Web Design:
 - AI-powered CSS generators (e.g., CSS Scan, Pika)
 - AI-based color palette tools (e.g., Khroma, Colormind)

1.7 Styling Text & AI-Driven Typography

- Font Properties (font-size, font-family, font-weight, line-height)
- Text Decoration & Alignment
- Using Google Fonts & AI font pairing tools (e.g., FontJoy)
- Implementing AI-generated typography for better readability

1.8 CSS Position Layout

- Static, Relative, Absolute, Sticky & Fixed
- Z-index
- Overflow
- Implementing AI-generated for Position

1.9 CSS Layouts & AI-Enhanced Design

- Display properties (block, inline, flex, grid)
- Box Model: margin, padding, border, width, height
- Flexbox & AI layout recommendations
- CSS Grid & AI-assisted responsive design tools
- AI-powered automatic layout generation (e.g., Figma's AI features)

1.10 Responsive Design & AI Optimization

- Media Queries & Viewport
- CSS Variables & Custom Properties
- AI-generated adaptive designs for different screen sizes
- AI-powered automated testing for responsive designs (e.g., Google Lighthouse, BrowserStack)
- Introduction to AI-driven CSS frameworks (e.g., Tailwind CSS with AI utilities)

1.11 CSS Animations, Transitions & AI Motion Design

- CSS Transitions & Transformations (rotate, scale, translate)
- Keyframe Animations (@keyframes)
- CSS Animation Libraries (e.g., Animate.css, GSAP)
- AI-generated animations for UI/UX enhancements
- AI-assisted motion design (e.g., Framer Motion AI)

1.12 Introduction to SASS/SCSS & AI in Styling

- What is SASS?
- Differences between SASS and SCSS
- Setting up a development environment
- Installing SASS via CLI & package managers
- AI in CSS Preprocessing:
 - AI-powered CSS generators
 - AI-based color scheme and design tools (Colormind, Khroma)
 - Variables, Nesting, & AI-Generated Styles
- Variables, Nesting
- Mixins, Functions & AI-Enhanced Code Efficiency

- Extending & Modularizing with AI Automation
- Advanced Features & AI-Powered Frameworks

1.13 UI Frameworks

- Bootstrap
 - Colors, Spacing, and Typography
 - Bootstrap Grid System
 - Bootstrap Components
 - Creating a custom Bootstrap theme
- Tailwind CSS
 - Colors, Spacing, and Typography
 - Backgrounds, Borders, and Shadows
 - Flexbox & Grid system
 - Responsive design principles
 - Transitions & Animations

1.14 JavaScript ES6+ Features & DOM Manipulation

- Variables (var, let, const)
- Data Types & Operators
- Loops, Functions, and Scope
- ES6 Features (Arrow Functions, Spread Operator, Destructuring)
- Hands-on: Building a Calculator with JavaScript

1.15 DOM Manipulation & Events

- Selecting Elements (querySelector, getElementById)
- Handling Events (click, input, keypress)
- Modifying HTML & CSS Dynamically
- Hands-on: Building a To-Do List App

1.16 JQuery

- Overview of jQuery and its advantages
- Basic jQuery syntax and selectors
- jQuery Events and Effects (show, hide, toggle, fade, slide etc)
- Using jQuery plugins

1.17 Asynchronous JavaScript (Fetch API, Promises, Async/Await)

- Callbacks & Promises

- Fetch API for HTTP Requests
- Async/Await for API Calls
- Hands-on: Fetching Data from an API and Displaying it on UI

1.18 Introduction to React

- JSX and Components
 - Creating functional and class components.
 - Understanding JSX syntax.
- State and Props
 - Managing component state.
 - Passing and validating props.
- Lifecycle Methods
 - Overview of React lifecycle phases.

1.19 React-Router-Dom

- Advanced routing concepts (dynamic routes, nested routes).

1.20 Form Handling

- Using Formik or React Hook Form for form validation.

1.21 Introduction to APIs

- Differences between REST, GRPC, SOAP, and GraphQL.

1.22 Introduction to Next.js

- Introduction to Next.js and its advantages
- Setting up a Next.js project
- Understanding project structure and file conventions

1.23 Styling in Next.js

- CSS Modules in Next.js
- Using Tailwind CSS with Next.js
- Styled Components and Emotion
- Global Styles and Theme Customization

1.24 Next.js Routing and Navigation

- Pages and File-based Routing
- Dynamic and Nested Routes

1.25 Fetching and Managing Data

- Fetching Data with fetch() and axios

- Server-side Rendering (SSR) vs Static Site Generation (SSG)
- Incremental Static Regeneration (ISR)

1.26 State Management in Next.js

- Context API and useReducer Hook

1.27 Performance Optimization

- Image Optimization with Next.js Image Component
- Lazy Loading and Code Splitting
- Optimizing API Calls and Caching
- SEO Best Practices in Next.js

1.28 AI Integration in React/Next.js

- Adding AI-powered features like chatbots or recommendation systems

1.29 Deployment and Hosting in React/Next.js

- Deploying Next.js on Vercel

1.30 Projects (As per the Institute Rules, all four of the following projects will be done by the Student themselves! On ReactJs / NextJs)

- E-commerce Website
- Corporate Website
- News Portal Website
- Travel & Tours Website

[Total Duration: 45 Hours at 1.5 Hour per Day]